

BAYESIAN MODELLING IN FINITE POPULATIONS

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Summary: In this paper, the prediction problem in finite populations is considered from a modelling point of view. Predictors for the population total and population variance, which take into consideration the possibility that each one out of a range of models may be the correct model, are proposed. Applications are made to normal superpopulation models.

1. INTRODUCTION

Model misspecification has always been a serious objection to the utilization of superpopulation models in finite populations. See for example Herson (1976) and Hansen et al. (1983). The approaches proposed (Bayesian or not) to counter such difficulties have always been to impose restrictions in the samples to be selected. Among important works on the subject it is worth mentioning Royall and Herson (1973), Pereira and Rodrigues (1983), Royall and Pfefferman (1982) and Bolfarine et al. (1987). But such restrictions typically lead to some loss of efficiency if the model is correct. The approach proposed in this paper, rather than imposing restrictions (sometimes unrealistically) on the samples to be selected, is based on the consideration of a range $M_1, \dots, M_k$ of models that might be plausible for the problem at hand. Then, we propose predictors which adaptively consider the possibility that each one of the alternative models

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$M_1, \dots, M_k$ is the correct model. The work was inspired on Smith's (1986) Bayesian thoughts on modelling and estimation of parameters in infinite populations. Related approaches are given by Box and Tiao (1962) and (1968).

The finite population is denoted by $\mathcal{P} = \{1, \dots, N\}$, where N is known. Associated with unit j , there are $M+1$ quantities, $y_j, x_{j1}, \dots, x_{jM}$, where all but y_j are known.

Relating the two sets of quantities we consider the linear model

$$(1) \quad \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$

where $E[\mathbf{e}] = 0$, $Var[\mathbf{e}] = \sigma^2 \mathbf{W}$, where $\mathbf{W}$ is known and diagonal and

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}; \mathbf{X} = \begin{pmatrix} \mathbf{X}'_1 \\ \vdots \\ \mathbf{X}'_N \end{pmatrix}; \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_M \end{pmatrix};$$

with $\mathbf{X}'_j = (x_{j1}, \dots, x_{jM})$.

To gain information about a function of $\mathbf{y}$ that might, for instance, be the population total, $T = \sum_{i=1}^N y_i$, or the population variance, $S_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / N$, where $\bar{y} = \sum_{i=1}^N y_i / N$, the population mean, a sample s of size n is selected. As in Rodrigues et al. (1985), let i_j equal 1 or 0 according to whether unit j belongs to s or not. Let $\mathbf{I}_s = \text{diag}(i_1, \dots, i_N)$. Then we may write

$$\mathbf{y}_1 = \mathbf{I}_s \mathbf{y}; \mathbf{y}_2 = (\mathbf{I} - \mathbf{I}_s) \mathbf{y}; \mathbf{X}_1 = \mathbf{I}_s \mathbf{X}; \mathbf{X}_2 = (\mathbf{I} - \mathbf{I}_s) \mathbf{X};$$

$$\mathbf{W}_1 = \mathbf{I}_s \mathbf{W}; \mathbf{W}_2 = (\mathbf{I} - \mathbf{I}_s) \mathbf{W},$$

where the subscript 1 corresponds to the observed units and subscript 2 corresponds to the units not in s .

In Section 2 of the present paper we consider the prediction problem in finite populations when a range of alternative models $M_1, \dots, M_k$ is under consideration. Predictors that adaptively consider the possibility that each one of the models M_i is the correct model are considered. In Section 3, applications are considered for normal superpopulation models. Numerical computations illustrate the behaviour of the suggested predictors.

2. THE MODELLING APPROACH TO FINITE POPULATION PREDICTION

As in Smith (1986) let $M_1, \dots, M_k$ be the range of models under consideration. Let

$$p(\mathbf{y}, \theta^{(j)} | M_j) = p(\mathbf{y} | \theta^{(j)} | M_j) p(\theta^{(j)} | M_j)$$

be the probabilistic model for the vector $\mathbf{y}$, M_j being the correct model and $\theta^{(j)}$ the vector of unknown parameters appearing in M_j , $j = 1, \dots, k$. Then, if $\pi_j = P(M_j)$ denotes the prior probability that M_j is the correct model, we have, corresponding to the observed sample $\mathbf{y}_1$,

$$(2) \quad \pi_j^* = \frac{p(\mathbf{y}_1 | M_j) \pi_j}{\sum_{j=1}^k p(\mathbf{y}_1 | M_j) \pi_j}, j = 1, \dots, k,$$

the posterior probability attached to the model M_j , where

$$p(\mathbf{y}_1 | M_j) = \int p(\mathbf{y}_1 | \theta^{(j)}, M_j) p(\theta^{(j)} | M_j) d\theta^{(j)}$$

is the predictive density corresponding to the observed units $\mathbf{y}_1$. Inference about $\theta^{(j)}$ is based on the posterior density

$$(3) \quad p(\theta^{(j)} | \mathbf{y}_1, M_j) \propto p(\mathbf{y}_1 | \theta^{(j)}, M_j) p(\theta^{(j)} | M_j).$$

Model (1) is now rewritten, under model M_j , as

$$(4) \quad \mathbf{y} = \mathbf{X}^{(j)} \beta^{(j)} + \mathbf{e}^{(j)}$$

where $E[\mathbf{e}^{(j)}] = 0$, $\text{Var}[\mathbf{e}^{(j)}] = \sigma^2 \mathbf{W}^{(j)}$, and $\mathbf{W}^{(j)}$ is known and diagonal, $j = 1, \dots, k$. Similarly, we extend the notation $\mathbf{X}_1, \mathbf{X}_2, \mathbf{W}_1$ and $\mathbf{W}_2$ under (1) to the corresponding notations $\mathbf{X}_1^{(j)}, \mathbf{X}_2^{(j)}, \mathbf{W}_1^{(j)}$ and $\mathbf{W}_2^{(j)}$ under (4).

An overall predictor of T is then given by

$$\begin{aligned}
 (5) \quad \hat{T}_B^* &= E[T|\mathbf{y}_1] = \sum_{j=1}^k \pi_j^* E[T|\mathbf{y}_1, M_j] \\
 &= n\bar{y}_1 + \sum_{j=1}^k \pi_j^* \mathbf{1}'_N \mathbf{X}_2^{(j)} E[\beta^{(j)}|\mathbf{y}_1, M_j]
 \end{aligned}$$

where $\bar{y}_1 = \sum_{i \in s} y_i/n$, is the sample mean, $\mathbf{1}_N$ is a vector of ones of dimension N and $E[\beta^{(j)}|\mathbf{y}_1, M_j]$ may be computed from (3). Note that expression (5) follows from the result

$$E[\mathbf{y}_2|\mathbf{y}_1, M_j] = E\{E[\mathbf{y}_2|\mathbf{y}_1, \beta^{(j)}, M_j]|\mathbf{y}_1, M_j\} = \mathbf{X}_2^{(j)} E[\beta^{(j)}|\mathbf{y}_1, M_j],$$

and from $T = \mathbf{1}'_N \mathbf{y}_1 + \mathbf{1}'_N \mathbf{y}_2$, where $\mathbf{1}'_N \mathbf{y}_1 = n\bar{y}_1$. Similarly, with respect to the population variance S_y^2 , we may write

$$(6) \quad \hat{S}_{yB}^{2*} = E[S_y^2|\mathbf{y}_1] = \sum_{j=1}^k \pi_j^* E[S_y^2|\mathbf{y}_1, M_j],$$

where, under normality, $E[S_y^2|\mathbf{y}_1, M_j]$ may be computed from (13) in Rodrigues et al. (1985) and π_j^* from (2) above.

3. THE NORMAL LINEAR SUPERPOPULATION MODEL

Suppose that in (4), under the model M_j , $\mathbf{e}^{(j)} \sim N(0, \sigma^2 \mathbf{W}^{(j)})$, $j = 1, \dots, k$. Assume also the following informative prior structure for $\theta^{(j)} = (\beta^{(j)}, \sigma^2)$:

$$\begin{aligned}
 (7) \quad p(\beta^{(j)}, \sigma^2 | M_j) &= p(\beta^{(j)} | \sigma^2) p(\sigma^2) \\
 &\propto \frac{1}{\sigma^{m_j}} \exp\{-h(\beta^{(j)} - \mathbf{b}_0^{(j)})'(\beta^{(j)} - \mathbf{b}_0)/2\sigma^2\} \left(\frac{1}{\sigma^2}\right)^{\frac{\nu_0-1}{2}} \exp\{-\nu_0 S_0^2/2\sigma^2\},
 \end{aligned}$$

where m_j is equal to the dimension of $\beta^{(j)}$ and h, ν_0, S_0^2 and $\mathbf{b}_0$ are all known. Therefore, under model M_j of (4), it follows that

$$\begin{aligned}
 p(\mathbf{y}_1, \beta^{(j)}, \sigma^2 | M_j) &= p(\mathbf{y}_1 | \beta^{(j)}, \sigma^2, M_j) p(\beta^{(j)} | \sigma^2, M_j) p(\sigma^2) \\
 &\propto \sigma^{-n} \exp\{-(\mathbf{y}_1 - \mathbf{X}_1^{(j)})' \beta^{(j)}\} \mathbf{W}_1^{(j)} (\mathbf{y}_1 - \mathbf{X}_1^{(j)} \beta^{(j)})/2\sigma^2\} \\
 &\quad \sigma^{-m_j} \exp\{-h(\beta^{(j)} - \mathbf{b}_0^{(j)})'(\beta^{(j)} - \mathbf{b}_0)/2\sigma^2\} \\
 &\quad (\sigma^{-2})^{\frac{\nu_0-1}{2}} \exp\{-\nu_0 S_0^2/2\sigma^2\}.
 \end{aligned}$$

By integrating out $\beta^{(j)}$ and σ^2 in (8), it can be shown that

$$(9) \quad p(\mathbf{y}_1 | M_j) = \frac{1}{|\mathbf{X}_1'^{(j)} \mathbf{W}_1^{(j)-1} \mathbf{X}_1^{(j)} + h \mathbf{I}^{(j)}|^{\frac{1}{2}} (\sqrt{2\pi})^{n-m_j} (S_d^{(j)})^{\frac{a_j}{2}-1}},$$

where $a_j = n + \nu_0 - m_k$, $\mathbf{I}^{(j)}$ is the identity matrix with m_j rows and

$$S_d^{(j)} = (\mathbf{y}_1 - \mathbf{X}_1^{(j)} \hat{\beta}^{(j)})' \mathbf{W}_1^{(j)-1} (\mathbf{y}_1 - \mathbf{X}_1^{(j)} \hat{\beta}^{(j)}) + \nu_0 S_0^2 + (\hat{\beta}^{(j)} - \mathbf{b}_0^{(j)})' \left\{ (\mathbf{X}_1'^{(j)} \mathbf{W}_1^{(j)} \mathbf{X}_1^{(j)})^{-1} + h^{-1} \mathbf{I}^{(j)} \right\}^{-1} (\hat{\beta}^{(j)} - \mathbf{b}_0^{(j)}),$$

where $\hat{\beta}^{(j)}$ is the least squares estimator of $\beta^{(j)}$ under model M_j . From (8) it also follows that the marginal posterior density of $\beta^{(j)}$ is

$$(10) \quad p(\beta^{(j)} | \mathbf{y}_1, M_j) \propto \left\{ 1 + \frac{1}{S_d^{(j)}} (\beta^{(j)} - \hat{\beta}^{(j)})' \left\{ \mathbf{X}_1'^{(j)} \mathbf{W}_1^{(j)} \mathbf{X}_1^{(j)} + h \mathbf{I}^{(j)} \right\} (\beta^{(j)} - \hat{\beta}^{(j)}) \right\}^{-\frac{(n-m_j)}{2}},$$

where

$$\hat{\beta}^{(j)} = (\mathbf{X}_1'^{(j)} \mathbf{W}_1^{(j)-1} \mathbf{X}_1^{(j)} + h \mathbf{I}^{(j)})^{-1} (\mathbf{X}_1'^{(j)} \mathbf{W}_1^{(j)-1} \mathbf{X}_1^{(j)} \hat{\beta}^{(j)} + h \mathbf{b}_0^{(j)}).$$

In the particular case where a noninformative prior distribution is considered for $(\beta^{(j)}, \sigma^2)$,

$$p(\beta^{(j)}, \sigma^2) \propto 1/\sigma,$$

it follows from (9) that

$$(11) \quad p(\mathbf{y}_1 | M_j) = \frac{1}{|\mathbf{X}_1'^{(j)} \mathbf{W}_1^{(j)-1} \mathbf{X}_1^{(j)}|^{\frac{1}{2}} (\sqrt{2\pi})^{n-m_j} (S_d^{(j)})^{\frac{a_j}{2}-1}},$$

where

$$S_d^{(j)} = (\mathbf{y}_1 - \mathbf{X}_1^{(j)} \hat{\beta}^{(j)})' \mathbf{W}_1^{(j)-1} (\mathbf{y}_1 - \mathbf{X}_1^{(j)} \hat{\beta}^{(j)})$$

and $a_j = n + 1 - m_j$, $j = 1, \dots, k$. The posterior density for $\beta^{(j)}$ can easily be obtained from (10).

EXAMPLE 1. Let

$$(12) \quad M_j : \mathbf{y}_i = \beta_0 \delta_{1j} + x_i \beta_1 \delta_{2j} + x_i^2 \beta_2 \delta_{3j} + e_i^{(j)},$$

$i = 1, \dots, N, j = 1, \dots, 4$, where $\mathbf{e}^{(j)} \sim N(0, \sigma^2 \mathbf{W}^{(j)})$,

$$\mathbf{W}^{(j)} = \text{diag}(x_1^{\delta_{2j}}, \dots, x_N^{\delta_{2j}}),$$

and

$$\delta_{1j} = \begin{cases} 1; & j = 1, 3, 4 \\ 0; & j = 2 \end{cases}; \delta_{2j} = \begin{cases} 1; & j = 2, 3, 4 \\ 0; & j = 1 \end{cases}; \delta_{3j} = \begin{cases} 1; & j = 4 \\ 0; & j = 1, 2, 3. \end{cases}$$

Considering noninformative prior distribution for (β, σ^2) , follows from (9) that

$$(13) \quad p(\mathbf{y}_1 | M_j) = C^{(j)} (S_d^{(j)})^{-\frac{(a_j-2)}{2}},$$

where $M^{(1)} = M^{(2)} = 1, M^{(3)} = 2, M^{(4)} = 3$ and

$$C^{(1)} = \frac{2^{\frac{n-2}{2}} \Gamma(\frac{n}{2} - 1)}{\sqrt{n} (2\pi)^{\frac{n-1}{2}}}, S_d^{(1)} = \sum_{i \in s} (y_i - \bar{y}_1)^2, a_1 = n;$$

$$C^{(2)} = \frac{2^{\frac{n-2}{2}} \Gamma(\frac{n}{2} - 1)}{\sqrt{\sum_{i \in s} x_i} (2\pi)^{\frac{n-1}{2}}}, S_d^{(2)} = \sum_{i \in s} \frac{1}{x_i} (y_i - \frac{\bar{y}_1}{\bar{x}_1} x_i)^2, a_2 = n;$$

$$C^{(3)} = \frac{2^{\frac{n-3}{2}} \Gamma(\frac{n-1}{2} - 1)}{\sqrt{D^{(3)}} (2\pi)^{\frac{n-2}{2}}}, S_d^{(3)} = \sum_{i \in s} \frac{1}{x_i} (y_i - \hat{\beta}_0^{(3)} - \hat{\beta}_1^{(3)} x_i)^2, a_3 = n - 1;$$

$$C^{(4)} = \frac{2^{\frac{(n-4)}{2}} \Gamma(\frac{n-2}{2} - 1)}{\sqrt{D^{(4)}} (2\pi)^{\frac{n-3}{2}}}, S_d^{(4)} = \sum_{i \in s} \frac{1}{x_i} (y_i - \hat{\beta}_0^{(4)} - \hat{\beta}_1^{(4)} x_i - \hat{\beta}_2^{(4)} x_i^2)^2, \\ a_4 = n - 2,$$

where

$$D^{(3)} = \sum_{i \in s} x_i^{-1} \sum_{i \in s} x_i - n^2,$$

$$D^{(4)} = \sum_{i \in s} x_i^{-1} \sum_{i \in s} x_i \sum_{i \in s} x_i^3 + 2n \sum_{i \in s} x_i \sum_{i \in s} x_i^2 - \left(\sum_{i \in s} x_i \right)^3 \\ - \left(\sum_{i \in s} x_i^2 \right)^2 \sum_{i \in s} x_i^{-1} - n^2 \sum_{i \in s} x_i^3$$

and $\hat{\beta}_i^{(j)}$ is the least squares estimator of $\beta_i^{(j)}$, $i = 0, 1, 2, j = 1, 2, 3, 4$. The Bayes predictor of $T = \sum_{i=1}^N y_i$, that adaptively considers each one of the possible models $M_1, \dots, M_4$ to be the correct model, follows from (5), with π_j^* following from (13) and $E[\beta^{(j)} \mathbf{y}_1, M_j] = \hat{\beta}^{(j)}$, the least squares estimator of $\beta^{(j)}$ under model M_j .

EXAMPLE 2. Suppose now that model M_j is such that

$$M_j : y_i = \beta_0 \delta_{1j} + \beta_1 x_i \delta_{2j} + e_i^{(j)},$$

$j = 1, 2$ and the $e_i^{(j)}$ are independent and $N(0, \sigma^2 \mathbf{W}^{(j)})$, where $\mathbf{W}^{(j)} = \text{diag}(x_1^{\delta_{2j}}, \dots, x_N^{\delta_{2j}})$,

$$\delta_j = \begin{cases} 1; & j = 1 \\ 0; & j = 2 \end{cases} \text{ and } \delta_{2j} = \begin{cases} 1; & j = 2 \\ 0; & j = 1 \end{cases}.$$

Therefore, a Bayesian predictor of S_y^2 , which takes into consideration the possibility that each one of M_1 and M_2 is the correct model is given by

$$\hat{S}_{yB}^* = \pi_1^* E[S_y^2 | \mathbf{y}_1, M_1] + \pi_2^* E[S_y^2 | \mathbf{y}_1, M_2],$$

where π^* follows from (13) and $E[S_y^2 | \mathbf{y}_1, M_i]$, $i = 1, 2$, are given in Rodrigues et al. (1985), Examples 1 and 2 of Section 4.

The standard normal deviates and the random numbers used in the simulation studies that follow, were generated by using the subroutines *GGNML* and *GGUBS*, both from the *IMSL* library.

SIMULATION 1. To illustrate the behavior of predictor (5) under generated populations, some numerical comparisons are made with the expansion and the ratio estimators. For several values of β_0 , populations of size $N = 100$ were generated according to the model $y_i = \beta_0 + 1.2x_i + e_i$, where the e_i are independent and $N(0, \sigma^2 x_i)$, $i = 1, \dots, N$ and $\sigma^2 = 1.0$.

The x_i were generated independently, $U(1; 10)$. The following range of alternative models was considered:

$$M_j : y_i = \beta_0 \delta_{1j} + x_i \beta_1 \delta_{2j} + e_i^{(j)},$$

$i = 1, \dots, N$ and $j = 1, 2, 3$, where $\text{Var}[e_j] = \sigma^2 x_i^{\delta_{2j}}$,

$$\delta_{1j} = \begin{cases} 1; & j = 1, 3 \\ 0; & j = 2 \end{cases} \text{ and } \delta_{2j} = \begin{cases} 1; & j = 2, 3 \\ 0; & j = 1 \end{cases}$$

As shown in Bolfarine et al. (1987), the ratio estimator $\hat{T}_R = \sum_{i=1}^N x_i \bar{y}_1 / \bar{x}_1$, where $\bar{y}_1 = \sum_{i \in S} y_i / n$ and $\bar{x}_1 = \sum_{i \in S} x_i / n$ are the sample means corresponding to y and x , respectively, is the Bayesian predictor of $T = \sum_{i=1}^N y_i$ under model M_2 , with noninformative prior on (β, σ^2) . $\hat{T}_R$ is also the optimal (unbiased and minimum variance) predictor under M_2 as follows from Rodrigues et al. (1985). Similar results hold for the expansion predictor, $\hat{T}_E = N\bar{y}_1$, under model M_1 .

From each generated population, 1000 simple random samples of size $n = 10$ were selected without replacement. The mean squared error (MSE) of $\hat{T}_E, \hat{T}_R$ and $\hat{T}_B^*$ of (5), were estimated by $\Sigma(\hat{T} - T)^2/1000$, where the summation extends over the 1000 selected samples. Table 1 below shows the result of the simulation study. Note that for $\beta_0 = 0$, $\hat{T}_R$ is the best predictor. On the other hand, as β_0 increases, $\hat{T}_B^*$ becomes more and more efficient (robust) in the sense that it is not affected by changes in β_0 .

TABLE 1

β_0	$\frac{MSE(\hat{T}_R)}{MSE(\hat{T}_E)}$	$\frac{MSE(\hat{T}_B^*)}{MSE(\hat{T}_E)}$
0	0,0712	0,0765
15	0,130	0,0771
20	0,170	0,0769

SIMULATION 2. In this case, a population of size $N = 100$ was generated according to the model

$$y_i = 0.5 + 0.7x_i + 0.01x_i^2 + e_i,$$

where the e_i are independent and $N(0, \sigma^2 \mathbf{x}_i)$, $i = 1, \dots, N$ where $\sigma^2 = 5.0$ and the $\mathbf{x}_i$ were generated as $U(1; 10)$. The alternative models considered are the models in (12) where the predictors $\hat{T}_R$, $\hat{T}_E$ and $\hat{T}_B^*$ considered are those in Simulation 1. As in that case, 1000 simple random samples of size $n = 10$ were selected and estimators for $MSE(\hat{T})$ were computed. The results are summarized in Table 2, which shows the efficiency of $\hat{T}_B^*$ with respect to the other two predictors.

TABLE 2

$\frac{MSE(\hat{T}_R)}{MSE(\hat{T}_E)}$	$\frac{MSE(\hat{T}_B^*)}{MSE(\hat{T}_E)}$
0.154	0.029

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